REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 12 Advanced | Module 22 Lesson 3 | Bohr's Model of the Atom

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Hydrogen: $E_n = -13.6/n^2$ eV; $r_n = (5.29 \times 10^{-11})n^2$ m; $E_{photon} = |E_f - E_i|$; $\lambda(nm) \approx 1240/E(eV)$.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 62–74.

P01. Rutherford's large-angle alpha-particle scattering implied a nucleus that is:

A. Spread uniformly through the entire atom B. Negatively charged and larger than the atom
C. Without mass D. Small dense and positively charged

Answer

D. Small dense and positively charged.

P02. In Bohr's model, permitted electron energies are:

A. Discrete B. Continuous at every radius
C. Always positive D. Independent of principal quantum number

Answer

A. Discrete.

P03. A photon is emitted when an electron:

A. Moves from lower to higher energy without external input B. Moves from higher to lower energy
C. Stays in the same stationary state D. Disappears without energy transfer

Answer

B. Moves from higher to lower energy.

P04. Negative hydrogen bound-state energy means the electron:

A. Has negative mass B. Cannot absorb a photon
C. Needs added energy to reach the zero-energy free state D. Has zero binding energy

Answer

C. Needs added energy to reach the zero-energy free state.

P05. The Balmer series of hydrogen ends at:

A. n = 1 B. n = 3
C. n = infinity D. n = 2

Answer

D. n = 2.

P06. Calculate the hydrogen energy level En for n = 3, using $E_n = -13.6/n^2$ eV.

A. -3.022 eV B. -0.7556 eV
C. -1.511 eV D. 1.511 eV

Answer

C. -1.511 eV. $E_n = -13.6/3^2 = -1.511$ eV. Negative energy denotes a bound electron.

P07. Find the hydrogen orbital radius for n = 3. Use $r_n = (5.29 \times 10^{-11})n^2$ m.

A. 9.52×10^{-10} m B. 2.38×10^{-10} m
C. -4.76×10^{-10} m D. 4.76×10^{-10} m

Answer

D. 4.76×10^{-10} m. $r_n = (5.29 \times 10^{-11})(3^2) = 4.76 \times 10^{-10}$ m.

P08. A hydrogen electron falls from n = 4 to n = 2. Find the emitted photon energy. Use $E_n = -13.6/n^2$ eV.

A. 2.55 eV B. 5.1 eV
C. 1.275 eV D. -2.55 eV

Answer

A. 2.55 eV. $E_{photon} = E_i - E_f = (-13.6/4^2) - (-13.6/2^2) = 2.55$ eV.

P09. Find the minimum energy needed to ionise hydrogen from n = 3, using $E_n = -13.6/n^2$ eV.

A. 3.022 eV B. 1.511 eV
C. 0.7556 eV D. -1.511 eV

Answer

B. 1.511 eV. Ionisation raises energy from En to 0: $\Delta E = 0 - (-13.6/3^2) = 1.511$ eV.

P10. An atomic transition emits a photon of energy 2.5 eV. Calculate its wavelength in nanometres using $hc \approx 1240$ eV nm.

A. 992 nm B. 248 nm
C. 496 nm D. -496 nm

Answer

C. 496 nm. $\lambda = hc/E = 1240/2.5 = 496$ nm.

P11. Distinguish ground state from excited state.

Answer

The ground state is the lowest allowed energy state; an excited state is a higher allowed bound energy state.

P12. Why are line spectra characteristic of an element?

Answer

Its allowed energy-level differences determine specific emitted and absorbed photon energies.

P13. What major problem did the classical planetary atom model face?

Answer

An accelerating orbiting charge would radiate energy and spiral inward, contrary to stable atoms, and the model could not explain discrete spectra correctly.

P14. What is the zero-energy reference in the simple hydrogen model?

Answer

A free electron at rest infinitely far from the proton, with no binding interaction energy remaining.

P15. State one limitation of the Bohr model.

Answer

It describes hydrogen-like single-electron systems well in limited respects but does not correctly predict the full energy structure of multi-electron atoms.

P16. Calculate the hydrogen energy level En for n = 4, using $E_n = -13.6/n^2$ eV.

Answer

$E_n = -13.6/4^2 = -0.85$ eV. Negative energy denotes a bound electron.

P17. Calculate the hydrogen energy level En for n = 5, using $E_n = -13.6/n^2$ eV.

Answer

$E_n = -13.6/5^2 = -0.544$ eV. Negative energy denotes a bound electron.

P18. Find the hydrogen orbital radius for n = 4. Use $r_n = (5.29 \times 10^{-11})n^2$ m.

Answer

$r_n = (5.29 \times 10^{-11})(4^2) = 8.46 \times 10^{-10}$ m.

P19. Find the hydrogen orbital radius for n = 5. Use $r_n = (5.29 \times 10^{-11})n^2$ m.

Answer

$r_n = (5.29 \times 10^{-11})(5^2) = 1.32 \times 10^{-9}$ m.

P20. A hydrogen electron falls from n = 5 to n = 2. Find the emitted photon energy. Use $E_n = -13.6/n^2$ eV.

Answer

$E_{photon} = E_i - E_f = (-13.6/5^2) - (-13.6/2^2) = 2.856$ eV.

P21. A hydrogen electron falls from n = 6 to n = 2. Find the emitted photon energy. Use $E_n = -13.6/n^2$ eV.

Answer

$E_{photon} = E_i - E_f = (-13.6/6^2) - (-13.6/2^2) = 3.022$ eV.

P22. Find the minimum energy needed to ionise hydrogen from n = 4, using $E_n = -13.6/n^2$ eV.

Answer

Ionisation raises energy from En to 0: $\Delta E = 0 - (-13.6/4^2) = 0.85$ eV.

P23. Find the minimum energy needed to ionise hydrogen from n = 5, using $E_n = -13.6/n^2$ eV.

Answer

Ionisation raises energy from En to 0: $\Delta E = 0 - (-13.6/5^2) = 0.544$ eV.

P24. An atomic transition emits a photon of energy 3 eV. Calculate its wavelength in nanometres using $hc \approx 1240$ eV nm.

Answer

$\lambda = hc/E = 1240/3 = 413.3$ nm.

P25. An atomic transition emits a photon of energy 3.5 eV. Calculate its wavelength in nanometres using $hc \approx 1240$ eV nm.

Answer

$\lambda = hc/E = 1240/3.5 = 354.3$ nm.

P26. For hydrogen, calculate the photon energy emitted for the n = 3 to n = 2 transition.

Answer

E3 = −13.6/9 = −1.511 eV; E2 = −13.6/4 = −3.400 eV. Ephoton = E3 − E2 = 1.889 eV.

P27. Why do many alpha particles pass through thin gold foil with little deflection while a few scatter strongly?

Answer

Most of the atom is relatively empty space; rare close encounters with the small dense positive nucleus cause large deflections.

P28. Distinguish emission and absorption line spectra.

Answer

Emission lines arise when excited atoms emit selected photon energies. Absorption lines arise when atoms remove photons matching allowed upward transitions from an incident spectrum.

P29. A photon energy is smaller than every available upward transition from a hydrogen atom's ground state. Can it excite that atom through a single bound-state transition?

Answer

No. A single absorbed photon must match an allowed energy difference within the relevant linewidth. An unmatched smaller energy cannot produce that bound-state excitation.

P30. Compare hydrogen orbital radius and binding-energy magnitude as n increases.

Answer

Bohr radius grows as n^2, while the energy magnitude falls as $1/n^2$. Higher bound states are larger and less tightly bound.